

From DAPHNE Descriptors to

DUNE-DAQ Trigger Primitives

Non-beam and low-energy PDS signatures

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April 20, 2026

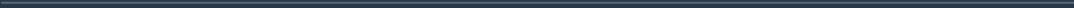




SECTION

Physics Context

What kind of light signatures matter, and what one board can or cannot decide



One-Slide Thesis

Physics signatures

For this trigger discussion, the relevant PDS cases are:

- ▶ prompt local light
- ▶ possible delayed local activity
- ▶ detector-wide low-light rate excess

Architecture consequence

One DAPHNE board only measures **channel-local** timing and pulse-shape observables. Firmware should export those descriptors so DAQ can build `TriggerPrimitive` objects and make detector-level decisions downstream.

Three Layers We Must Not Mix

Physics in LAr

Expected footprint:

- ▶ local flash
- ▶ delayed local activity
- ▶ sparse detector-wide burst

Board firmware

What one board measures:

- ▶ timestamp
- ▶ width, peak, integral
- ▶ peak count, quality bits

DAQ trigger

What the detector decides:

- ▶ TP, TA, TC, TD
- ▶ clustering across boards
- ▶ long time-window logic

Working rule

signature in LAr → descriptor in firmware → trigger object in DAQ

What The DUNE Literature Supports

Claim 1: PDS is not only for beam timing

DUNE PDS is explicitly relevant for non-beam triggering, t_0 , supernova sensitivity, and nucleon-decay-motivated searches.

Claim 2: Supernova is a rate problem

The useful signature is not one bright flash; it is a detector-wide excess of many low-light interactions over ms to s windows.

Claim 3: Very-low-energy light is noise-limited

Standalone optical triggering near threshold remains difficult; baseline stability and noise rejection are still central.



SECTION

Firmware to DAQ

What DAPHNE exports, what DAQ already defines, and what we need to validate



What One DAPHNE Board Can Export

Timing and trigger context

trigger_timestamp, baseline, filtered samples, and trigger_pulse.

Pulse descriptors

time_start, time_peak, time_over_baseline, adc_peak, adc_integral, number_peaks, and quality bits.

Important limitation

The public boundary still exports `TRIGGER_DESCRIPTOR_NULL`; the TP-facing export contract is not the finished firmware boundary yet.

Descriptors to DAQ Trigger Primitives

Direct mappings

<code>time_peak</code>	→	<code>samples_to_peak</code>
<code>time_over_baseline</code>	→	<code>samples_over_threshold</code>
<code>adc_peak</code>	→	<code>adc_peak</code>
<code>adc_integral</code>	→	<code>adc_integral</code>

waveform timestamp + sample offset → TP start timestamp

Reading

Firmware fills the local pulse quantities; DAQ instantiates the common `TriggerPrimitive` object.

What The DAQ Contract Already Defines

Already defined downstream

- ▶ TP bit widths and quality semantics
- ▶ channel and detector mapping path via `detchannelmaps`
- ▶ extended metadata path via `fddetdataformats`
- ▶ DAQ object model in `trigger` and `triggeralgs`

Next interface task

Validate the concrete DAPHNE-to-TP mapping against the existing DAQ object and metadata contracts.

The Trigger Hierarchy Is Downstream

Object flow

DAPHNE descriptor → DAQ TriggerPrimitive → TriggerActivity
TriggerCandidate → TriggerDecision

Firmware

- ▶ channel-local pulse finding
- ▶ peak descriptors
- ▶ precise timing context

DAQ object building

- ▶ common TP schema
- ▶ TA clustering inside a TDAQ unit
- ▶ TC merging across units

Physics decision

- ▶ burst logic
- ▶ delayed association
- ▶ PDS/TPC coincidence

Which Layer Should Decide What?

Board / TP level

- ▶ localized prompt light
- ▶ width, peak, integral
- ▶ low-quality hit tagging

TA / TC level

- ▶ delayed local association
- ▶ multi-channel clustering
- ▶ PDS/TPC coincidence

Detector level

- ▶ supernova burst logic
- ▶ multi-board multiplicity
- ▶ global trigger decision

Rule of thumb

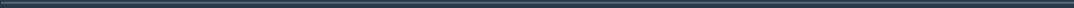
If the question needs detector footprint or long time-window context, it is already beyond one board.



SECTION

Scaling and Limits

The 40-channel board is local; the 6k-channel detector trigger is hierarchical



Scaling From One Board To ~ 6000 Channels

Current board scale

- ▶ `AFE_COUNT_G = 5`
- ▶ 8 channels per AFE
- ▶ 40 channels per board-equivalent shell

Detector scale

$$\frac{6000}{40} \approx 150$$

- ▶ about 150 board-equivalents

Implication

- ▶ keep descriptor production channel-local
- ▶ populate one stable DAQ TP schema
- ▶ build multiplicity above the board
- ▶ include busy/full/packet loss accounting

Current DAPHNE Trigger Dead Time

Context

Why it matters

Rate-limited descriptor export means TP logic must preserve loss accounting.

Key numbers

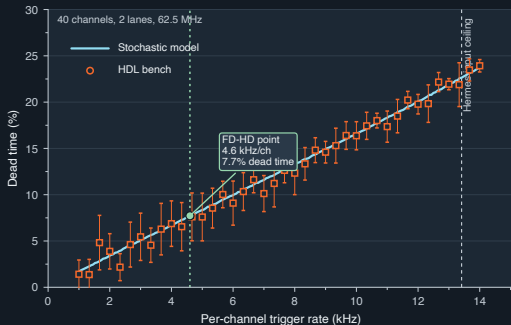
40 channels, 2 lanes, 62.5 MHz

$\tau_b = 16.592 \mu\text{s}$

$\mu_{\text{ch}} \approx 13.41 \text{ kHz/ch}$

1.68 % @ 1 kHz/ch

23.76 % @ 14 kHz/ch



What The Current Interface Already Covers

Already in place

- ▶ TP timestamp is waveform timestamp plus sample offset
- ▶ channel and detector IDs follow `detchannelmaps`
- ▶ TP widths, flags, and saturation semantics come from the DAQ TP definition
- ▶ trailer words survive as extended metadata through `fddetdataformats`

What still matters on our side

Validate the DAPHNE-to-TP field content, exercise PDS-oriented TA building, and check rate and loss behavior against board dead-time limits.

Takeaways

- ▶ DAPHNE should preserve the **local optical information** that is expensive to recover later: time, width, peak, integral, and quality.
- ▶ The correct common language downstream is the existing DUNE-DAQ `TriggerPrimitive` schema.
- ▶ Detector-scale signatures such as supernova bursts are inherently **multi-board and multi-timescale** problems.
- ▶ Keep four layers separate: physics signature, firmware descriptor, DAQ trigger object, and detector-level trigger decision.

Sources

- ▶ DUNE Collaboration, *ProtoDUNE-DP scintillation light detection*, 2022.
- ▶ Cattadori for DUNE, *FD1 X-ARAPUCA efficiency and FD2 photon collector upgrade*, 2024.
- ▶ Brizzolari for DUNE, *Features and performances of the DUNE Far Detector PDS*, 2024/2025.
- ▶ DUNE Collaboration, *Far Detector Vertical Drift TDR*, 2024.
- ▶ Andringa et al., *Low-energy physics in neutrino LArTPCs*, 2023.
- ▶ DUNE-DAQ `trigger`, `triggeralgs`, `detchannelmaps`, and `fddetdataformats`.